

# Juniper Networks<sup>®</sup> vMX Release Notes

Release 18.2R1

21 July, 2020

These release notes accompany this release of the Juniper Networks<sup>®</sup> virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

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# Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 18.2 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

## New and Changed Features

- Starting with Junos OS Release 18.2R1, you can bypass the queuing-chip on vMX routers to save vCPU resources when scheduling is not required on an interface.

You can use the **bypass-queuing-chip** option at the **[edit class-of-service interfaces interface-name]** hierarchy level to bypass the queuing-chip. When using the bypass queuing-chip option, we recommend you not to oversubscribe the interface bandwidth.

[See [Bypassing the Queuing Chip](#)].

## Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

# Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge. PR1059231
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- When the FPC is restarting, **COS(cos\_chassis\_scheduler\_pre\_add\_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- When the FPC is restarting, a **Received unsupported pic\_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- When committing the configuration, **%DAEMON-4-SNMP\_EVLIB\_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902
- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets\_reply:3545 opackets\_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync\_update\_tod\_to\_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344
- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK\_SCHED\_SLIP** log messages. PR1153290
- Observing **rpdd[7116]: %DAEMON-6: ifl\_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096

## Subscriber Management

- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154

## Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

## System Requirements

### IN THIS SECTION

- [Minimum Hardware and Software Requirements for KVM | 4](#)
- [Minimum Hardware and Software Requirements for VMware | 7](#)

These topics provide system requirements for each supported environment.

## Minimum Hardware and Software Requirements for KVM

[Table 1 on page 5](#) lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

| Description                                                                                                                 | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sample system configuration                                                                                                 | <p>For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability.</p> <p>For all other use cases, Intel Ivy Bridge processors or later are required.<br/>Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache</p> <p>For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Number of cores</p> <p><b>NOTE:</b> Performance mode is the default mode and the minimum value is based on one port.</p> | <p>For lite mode with lab simulation use case applications: Minimum of 3</p> <ul style="list-style-type: none"> <li>• 1 for VCP</li> <li>• 2 for VFP</li> </ul> <p><b>NOTE:</b> If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                             | <p>For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9</p> <ul style="list-style-type: none"> <li>• 1 for VCP</li> <li>• 8 for VFP</li> </ul> <p>The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP.</p> <p><b>NOTE:</b> To calculate the optimal number of vCPUs needed by VFP for performance mode:</p> <ul style="list-style-type: none"> <li>• Without CoS—<math>(4 * \text{number-of-ports}) + 4</math></li> <li>• With CoS—<math>(5 * \text{number-of-ports}) + 4</math></li> </ul> <p><b>NOTE:</b> All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for host OS on any server running the vMX.</p> |

Table 1: Minimum Hardware Requirements for KVM (*continued*)

| Description                                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Memory                                             | For lite mode: Minimum of 3 GB                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>NOTE:</b> Performance mode is the default mode. | <ul style="list-style-type: none"> <li>• 1 GB for VCP</li> <li>• 2 GB for VFP</li> </ul> For performance mode: <ul style="list-style-type: none"> <li>• Minimum of 5 GB               <ul style="list-style-type: none"> <li>1 GB for VCP</li> <li>4 GB for VFP</li> </ul> </li> <li>• Recommended of 16 GB               <ul style="list-style-type: none"> <li>4 GB for VCP</li> <li>12 GB for VFP</li> </ul> </li> </ul> Additional 2 GB recommended for host OS |
| Storage                                            | Local or NAS<br><br>Each vMX instance requires 30 GB of disk storage                                                                                                                                                                                                                                                                                                                                                                                                |
| Other requirements                                 | Intel VT-d capability<br><br>Hyperthreading (recommended)<br><br>AES-NI                                                                                                                                                                                                                                                                                                                                                                                             |

Table 2 on page 6 lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

| Description                                                                                                    | Value                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating system                                                                                               | Ubuntu 16.04.5 LTS (recommended host OS)<br>Linux 4.4.0-generic                                                                                                                                                                                                                            |
| Virtualization                                                                                                 | QEMU-KVM 2.0.0+dfsg-2ubuntu1.11                                                                                                                                                                                                                                                            |
| Required packages<br><br><b>NOTE:</b> Other additional packages might be required to satisfy all dependencies. | bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server<br>libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev<br>libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip<br>python-dev libxml2-dev libxslt-dev<br><br><b>NOTE:</b> libvirt 1.3.1 |

[Table 3 on page 7](#) lists the software requirements for Red Hat Enterprise Linux.

**Table 3: Software Requirements for Red Hat Enterprise Linux**

| Description                                                                            | Value                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating system                                                                       | <ul style="list-style-type: none"> <li>Red Hat Enterprise Linux 7.2<br/>Linux 3.10.0-327.4.5</li> <li>Starting with Junos OS Release 17.3R1<br/>Red Hat Enterprise Linux 7.3<br/>Linux 3.10.0-514.6.2</li> </ul>                                                                                        |
| Virtualization                                                                         | QEMU-KVM 1.5.3                                                                                                                                                                                                                                                                                          |
| Required packages<br><br><b>NOTE:</b> SR-IOV requires these packages: kernel-devel gcc | python27-python-pip python27-python-devel numactl-libs libpciaccess-devel<br>parted-develyajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel<br>libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm<br>numactl telnet net-tools<br><br><b>NOTE:</b> libvirt 1.2.17 or later |

[Table 4 on page 7](#) lists the software requirements for CentOS.

**Table 4: Software Requirements for CentOS**

| Description       | Value                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating system  | CentOS 7.2<br>Linux 3.10.0-327.22.2                                                                                                                                                                                                                                                                                                                                                           |
| Virtualization    | QEMU-KVM 1.5.3                                                                                                                                                                                                                                                                                                                                                                                |
| Required packages | python27-python-pip python27-python-devel numactl-libs libpciaccess-devel<br>parted-develyajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel<br>libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl<br>telnet net-tools<br><br><b>NOTE:</b> libvirt 1.2.19<br><br>To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17. |

## Minimum Hardware and Software Requirements for VMware

[Table 5 on page 8](#) lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

| Description                                                                                                                 | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Number of cores</p> <p><b>NOTE:</b> Performance mode is the default mode and the minimum value is based on one port.</p> | <p>For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications:<br/>Minimum of 9</p> <ul style="list-style-type: none"> <li>• 1 for VCP</li> <li>• 8 for VFP</li> </ul> <p>The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP.</p> <p><b>NOTE:</b> To calculate the optimal number of vCPUs needed by VFP for performance mode:</p> <ul style="list-style-type: none"> <li>• Without CoS—<math>(4 * \text{number-of-ports}) + 4</math></li> <li>• With CoS—<math>(5 * \text{number-of-ports}) + 4</math></li> </ul> <p><b>NOTE:</b> All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.</p> <hr/> <p>For lite mode: Minimum of 3</p> <ul style="list-style-type: none"> <li>• 1 for VCP</li> <li>• 2 for VFP</li> </ul> <p><b>NOTE:</b> If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.</p> |
| <p>Memory</p> <p><b>NOTE:</b> Performance mode is the default mode.</p>                                                     | <p>For performance mode:</p> <ul style="list-style-type: none"> <li>• Minimum of 5 GB <ul style="list-style-type: none"> <li>1 GB for VCP</li> <li>4 GB for VFP</li> </ul> </li> <li>• Recommended of 16 GB <ul style="list-style-type: none"> <li>4 GB for VCP</li> <li>12 GB for VFP</li> </ul> </li> </ul> <p>For lite mode: Minimum of 3 GB</p> <ul style="list-style-type: none"> <li>• 1 GB for VCP</li> <li>• 2 GB for VFP</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



**Table 5: Minimum Hardware Requirements for VMware (*continued*)**

| Description | Value                                                                |
|-------------|----------------------------------------------------------------------|
| Storage     | Local or NAS<br><br>Each vMX instance requires 30 GB of disk storage |

[Table 6 on page 9](#) lists the software requirements.

**Table 6: Software Requirements for VMware**

| Description       | Value                         |
|-------------------|-------------------------------|
| Hypervisor        | ESXi 5.5 Update 2             |
| Management Client | vSphere 5.5 or vCenter Server |

# Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf>.
- Product warranties—For product warranty information, visit <http://www.juniper.net/support/warranty/>.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

## Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <https://prsearch.juniper.net/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://kb.juniper.net/InfoCenter/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Create a service request online: <https://myjuniper.juniper.net>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://entitlementsearch.juniper.net/entitlementsearch/>

## Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit <https://myjuniper.juniper.net>.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://support.juniper.net/support/requesting-support/>.

## Revision History

21 July 2020—Revision 5, Junos OS Release 18.2R1—vMX

21 February 2019—Revision 4, Junos OS Release 18.2R1—vMX

12 February 2019—Revision 3, Junos OS Release 18.2R1—vMX

11 October 2018—Revision 2, Junos OS Release 18.2R1—vMX

29 June 2018—Revision 1, Junos OS Release 18.2R1—vMX

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